

ST312 Summer 20/21

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Q.1

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P.1

- a) Experimental study: participants are assigned to control and experimental groups and the effects of the intervention are analyzed
- b) Response variable: skin health after one month
Explanatory variables: initial skin health
Treatment: short wave UVA-blocking sunscreen, long wave UVA-blocking sunscreen
Experimental units: individual participants
- c) The results are generalisable, as the sample was chosen randomly from the population and it should therefore be representative of it.
- d) Subjects were assigned at random because:
- this avoids bias, helps ensure both groups are comparable and that differences are due to the treatment
 - allows for causal inferences to be made.
 - random assignment can be used to justify probabilistic analysis and allows for randomisation tests to be carried out to gather evidence of the treatment's effectiveness
- e) Yes, if the participants are randomly assigned there should be a similar baseline on average between the control and treatment groups and the changes, or lack thereof, between baseline and post-intervention measurements can be attributed to the treatment.
Adherence to the treatment would be hard to gauge, however.
- f) The study could use a cross over design where participants assigned to the short wave UVA group later wear the long wave UVA-blocking sunscreen after a washout period, and vice versa for the initial long wave UVA control group.
Data could be measured more often rather than just twice
Data could be gathered about the weather that the sunscreens are being worn in

Q.2

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- (a) Factor of interest: type of corn
Nuisance factor: field
- (b) Experimental units: individual subplots within each field
- (c) Yes, the 10 fields are blocks
- (d) The experimenters aimed to use the randomised block design to control nuisance factors, hoping to balance out any variants between fields and reduce the error variability for corn yield comparisons within blocks.
- (e) Yes, there is evident field to field variation. Failing to account for this may make it difficult to draw conclusions about corn differences.
- (f) H_0 : no treatment differences $\mu_A = \mu_B = \mu_C$
 H_1 : at least one of the μ 's is different

Steps:

1. Shuffle results across corn within each field
2. Calculate F-statistic
3. Repeat 1000 times to give randomisation distribution under H_0
4. Calculate observed F-statistic and calculate p-value as proportion of randomised values $\geq$ observed
5. If p-value is below test significance level reject H_0

Can be used when the assumptions expected for a standard ANOVA F-test are not met

3_{ps} $H_0: \mu_{\text{postgrad}} = \mu_{\text{undergrad}}$

μ = mean exercise per week

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$H_1: \mu_{\text{postgrad}} \neq \mu_{\text{undergrad}}$

$$F\text{-statistic} = \frac{MS_{\text{groups}}}{MSE} = \frac{182.12}{10.23} \approx 17.80254$$

Since F is $> 3.92 (F_{1,120})$ we can reject H_0 .

This is evidence to suggest there is a significant difference between the mean weekly exercise of postgraduate and undergraduate students

(b) $t\text{-test statistic} = \sqrt{17.80254} \approx 4.21931$

(c) $H_0: \mu_{\text{No smoking}} = \mu_{\text{Sometimes Smoking}} = \mu_{\text{Yes smoking}}$

$H_1: \mu_i \neq \mu_j \text{ for some } i, j$

Test-statistic F -value is 4.327 and p -value is 0.0143. Since $p < 0.05$, we can reject H_0 and conclude there is some mean difference of students' weekly exercise depending on smoking habits

(d) Both the histogram and the Normal Q-Q Plot can be used to assess the normality of residuals assumption. The residuals v.s. fitted plot can be used to assess constant variance of residuals homoscedasticity assumption.

The Normal Q-Q Plot shows the observed roughly lying along the diagonal line, while there is some deviance in the tails there is not strong evidence of departure. This, along with the histogram suggests the normality assumption is valid

The pattern exhibited by the Residuals v.s. Fitted plot indicates heteroscedasticity in the data, the variance of the residuals is decreasing rather than the randomly scattered constant variance required by our assumption of homoscedasticity.

I suggest transforming the data (e.g. log transformation) and re-assessing the variance of residuals, or performing a non-parametric test (e.g. Kruskal-Wallis)

(c) Since H_0 was rejected, there are differences between the exercise habits of different types of smokers, we now investigate how they differ

(b) There is evidence that both non-smoking and sometimes-smoking students exercise more than smoking students, on average. We see this because the Yes-No and Yes-Sometimes 95% Confidence Intervals are negative. On the other hand, the Sometimes-No confidence interval contains 0, suggesting we cannot rule out no difference in mean weekly exercise between sometime-smoking and non-smoking students

$$(g) H_0: \mu_{no} - \frac{\mu_{yes} + \mu_{sometimes}}{2} = 0$$

$$H_1: \mu_{no} - \frac{\mu_{yes} + \mu_{sometimes}}{2} \neq 0$$

$$6 - \frac{(3.36 + 5.99)}{2} = 1.325 \quad \text{mean difference in weekly exercise of never smokers and ever smokers}$$

$$SE(\text{contrast}) = \sqrt{MSE \sum_{i=1}^k \frac{c_i^2}{n_i}}$$

$$= \sqrt{10.64 \times \left(\frac{1}{109} + \frac{(1/2)^2}{14} + \frac{(1/2)^2}{41} \right)} = 0.55783$$

$$\text{Test statistic} = \frac{1.325}{0.55783} = 2.3752$$

Using the t-tables, the critical values for $t_{242, 0.025}$ are approximately -1.98 and 1.98

Since our test statistic is higher than 1.98 we reject H_0

There is evidence that the mean weekly exercise is higher for never smoker students than ever smoker students

(d) Here the variation in university level and smoking habits are accounted for simultaneously, the variation in response is broken down into that explained by student and smoke groups. Here, both main effects remain significant when allowing for variation in the other.

(i) Since the lines are not parallel, this suggests an interaction might prove useful.

(j) Since the p value is large ($p = 0.5882$) we conclude there is no strong evidence for an interaction. The overall conclusion is that mean exercise is related to both level of study and smoking habits of students. In particular, students who never smoke and students who only sometimes smoke do more exercise per week on average than students who smoke, and undergraduate students exercise more often than postgraduate students, on average.

Q.4

	Ate	Did not
Hepatitis	18	1
Healthy	7	31

(i) Equality of proportions:

$$H_0: \pi_{\text{ate}} - \pi_{\text{didn't eat}} = 0, H_1: \pi_{\text{ate}} - \pi_{\text{didn't eat}} > 0$$

$$T.S. = \frac{18}{25} - \frac{1}{32} = 0.68875$$

(ii) Relative risk:

$$H_0: RR = \frac{\pi_{\text{ate}}}{\pi_{\text{didn't eat}}} = 1, H_1: RR = \frac{\pi_{\text{ate}}}{\pi_{\text{didn't eat}}} > 1$$

$$RR = \frac{18/25}{1/32} = 23.04$$

(iii) Odds-ratio

$$H_0: OR = \frac{\frac{\pi_{\text{ate}}}{1-\pi_{\text{ate}}}}{\frac{\pi_{\text{didn't eat}}}{1-\pi_{\text{didn't eat}}}} = 1, H_1: OR = \frac{\frac{\pi_{\text{ate}}}{1-\pi_{\text{ate}}}}{\frac{\pi_{\text{didn't eat}}}{1-\pi_{\text{didn't eat}}}} > 1$$

$$OR = \frac{18/7}{1/31} = \frac{(18)(31)}{(7)(1)} = 79.714$$

(b)

	Ate at Ron's	Didn't Eat at Ron's	Total
Hepatitis	19	0	19
Healthy	6	32	38
Total	25	32	57

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Fisher's p-value:

$$\frac{\binom{25}{18} \times \binom{32}{1}}{\binom{57}{19}} + \frac{\binom{25}{19} \times \binom{32}{0}}{\binom{57}{19}}$$

$$p \approx 2.418 \times 10^{-8} + 2.7803 \times 10^{-10} = 2.445803 \times 10^{-8}$$

Since the p-value is small (e.g. < 0.05), we can reject the null hypothesis that there is no association between Ron's grill and Hepatitis Cases. This suggests there is an association, with an increased risk in those that ate at Ron's grill during the past two months.

Q.5

(a) Logit = $-0.785102 + 0.012081$ (Contributions)

$$\text{Prob} = \frac{e^{-0.785102 + 0.012081 \text{ (Contributions)}}}{1 + e^{-0.785102 + 0.012081 \text{ (Contributions)}}}$$

(b) The slope for Contributions is 0.012081

For every \$1,000 contribution received from the pharmaceutical industry over 6 years, the log-odds of senators voting with the bill and keep drug prices high increases by 0.012081.

$$\text{The odds ratio} = e^{0.012081} = 1.01215$$

Odds of voting with increase by 1.01215 times for every \$1,000 contribution. For every contribution, the odds of voting to keep drug prices high increases by 1.25%.

(c) $H_0: \beta_1 = 0$, $H_1: \beta_1 \neq 0$

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P.7.

$$\text{Test statistic} = \frac{\beta_1}{SE(\beta_1)} = \frac{0.012081}{0.003617} = 3.34$$

p-value = 0.000838, since this is less than 0.05, we can reject H_0 and conclude that Contributions is a useful predictor of voting.

(d) Prob of voting with the bill receiving \$80,000 = $\frac{e^{-0.785102 + 0.012081(80)}}{1 + e^{-0.785102 + 0.012081(80)}} = 0.5452$

Probability of voting against = $1 - 0.5452 = 0.4548$

(e) 95% Confidence Interval for slope is : $0.012081 \pm 1.96 \times 0.003617$
0.00499, 0.019 on the log-odds scale

95% Confidence Interval for odds ratio is : $e^{0.00499}, e^{0.019}$
1.005, 1.019

With 95% confidence, the true population odds ratio is likely between 1.019 and 1.005. Since 1 is not in the interval, this suggests there is a significant association between contributions and voting, the odds likely increase by between 0.5% and 1.9% for every \$1,000 contribution.

(g) OR for males: $e^{0.016324} = 1.016 \dots$ 1.6% increased odds per day of drink for male students
OR for females: $e^{0.016324 - 0.0051} = 1.011 \dots$ 1.1% increased odds per day of drink for female students

